

Practical 1

Aim:

Linux System Administration and File Management for the Marketing Analytics Team.

Problem Definition:

Scenario: As a newly joined Linux System Administrator at DataAnalytics Inc., the company's analytics platform runs on Linux servers where project data, reports, scripts, and user accounts must be managed efficiently. The manager has assigned the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data, and creating backups for the Marketing Analytics team.

Objectives:

To perform Linux system administration and file management operations including user identification, directory navigation, system monitoring, directory/file creation, content handling, searching, and backup operations using Linux terminal commands.

Observation / Output (Screenshots or command outputs)

1. Identify the current user.

```
whoami  
id
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop$ whoami  
aubaid-ahmed
```

2. Find the current working directory.

```
pwd
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop$ pwd  
/home/aubaid-ahmed/Desktop
```

3. Display system information.

```
uname -a
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop$ uname -a  
Linux Ubuntu-Coursework 7.0.0-27-generic #27-Ubuntu SMP PREEMPT_DYNAMIC Thu Jun 18 19:13:49 UTC 2026 x86_64 GNU/Linux
```

4. Check system uptime and date.

uptime
date

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop$ uptime
21:58:47 up 22 min,  1 user,  load average: 0.42, 0.95, 0.83
aubaid-ahmed@Ubuntu-Coursework:~/Desktop$ date
Wed Aug  5 09:58:50 PM IST 2026
```

5. Create a directory named MarketingAnalytics.

```
mkdir MarketingAnalytics
cd MarketingAnalytics
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop$ mkdir marketingAnalytics
mkdir: marketingAnalytics: File exists
aubaid-ahmed@Ubuntu-Coursework:~/Desktop$ cd marketingAnalytics
```

6. Create subdirectories: RawData, ProcessedData, Reports.

```
mkdir RawData ProcessedData Reports
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ mkdir RawData ProcessedData Reports
mkdir: RawData: File exists
mkdir: ProcessedData: File exists
mkdir: Reports: File exists
```

7. Create data and report files.

```
touch RawData/customer_data.txt
touch Reports/sales_report.txt
touch Reports/marketing_report.txt
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ touch RawData/customer_data.txt
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ touch Reports/sales_report.txt
touch Reports/marketing_report.txt
```

8. Add sample content.

```
echo "CustomerID,Name,City" > RawData/customer_data.txt
echo "101,Raj,Ahmedabad" >> RawData/customer_data.txt
echo "102,Priya,Vadodara" >> RawData/customer_data.txt
```

```
echo "103,Amit,Surat" >> RawData/customer_data.txt
echo "Monthly Sales Report - Q1" > Reports/sales_report.txt
echo "Marketing Campaign Report" > Reports/marketing_report.txt
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ echo "CustomerID,Name,City" > RawData/customer_data.txt
echo "101,Raj,Ahmedabad" >> RawData/customer_data.txt
echo "102,Priya,Vadodara" >> RawData/customer_data.txt
echo "103,Amit,Surat" >> RawData/customer_data.txt

echo "Monthly Sales Report - Q1" > Reports/sales_report.txt
echo "Marketing Campaign Report" > Reports/marketing_report.txt
```

9. Display file contents.

```
cat RawData/customer_data.txt
cat Reports/sales_report.txt
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ cat RawData/customer_data.txt
cat Reports/sales_report.txt
CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat
Monthly Sales Report - Q1
```

10. List all files and directories.

```
ls -la
```

```
ls -R
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ ls -la
ls -R
total 28
drwxrwxr-x 7 aubaid-ahmed aubaid-ahmed 4096 Aug  5 21:55 .
drwxr-xr-x 3 aubaid-ahmed aubaid-ahmed 4096 Aug  5 21:43 ..
drwxrwxr-x 2 aubaid-ahmed aubaid-ahmed 4096 Aug  5 21:54 Backup
drwxrwxr-x 6 aubaid-ahmed aubaid-ahmed 4096 Aug  5 21:55 MarketingAnalytics
drwxrwxr-x 2 aubaid-ahmed aubaid-ahmed 4096 Aug  5 21:54 ProcessedData
drwxrwxr-x 2 aubaid-ahmed aubaid-ahmed 4096 Aug  5 21:59 RawData
drwxrwxr-x 2 aubaid-ahmed aubaid-ahmed 4096 Aug  5 21:59 Reports
.:
Backup MarketingAnalytics ProcessedData RawData Reports

./Backup:
```

11. Search for report files.

```
find . -name "*report*"
find . -type f -iname "*Report*"
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ find . -name "*report*"
find . -type f -iname "*Report*"
./Backup/sales_report.txt
./Reports/sales_report.txt
./Reports/marketing_report.txt
./Reports/monthly_sales_report.txt
[1]+  Exit 127                  *Report*
./Backup/sales_report.txt
./Reports/sales_report.txt
./Reports/marketing_report.txt
./Reports/monthly_sales_report.txt
```

12. Search specific content inside files.

```
grep "Vadodara" RawData/customer_data.txt
grep -r "Report" Reports/
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ grep "Vadodara" RawData/customer_data.txt
grep -r "Report" Reports/
102,Priya,Vadodara
Reports/sales_report.txt:Monthly Sales Report - Q1
Reports/marketing_report.txt:Marketing Campaign Report
Reports/monthly_sales_report.txt:Monthly Sales Report - Q1
```

13. Copy report files to Backup directory.

```
mkdir Backup
cp Reports/*.txt Backup/
```

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ mkdir Backup
mkdir: Backup: File exists
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ cp Reports/*.txt Backup/
```

14. Rename files.

```
mv Reports/sales_report.txt Reports/monthly_sales_report.txt
```

15. Move files between directories.

```
mv RawData/customer_data.txt ProcessedData/
```

16. Delete a file.

```
rm Backup/marketing_report.txt
```

17. Delete an empty directory.

```
mkdir TempDir  
rmdir TempDir
```

OUTPUT:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ mv Reports/sales_report.txt Reports/monthly_sales_report.txt  
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ mv RawData/customer_data.txt ProcessedData/  
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ rm Backup/marketing_report.txt  
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ mkdir TempDir  
rmdir TempDir  
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$
```

Result:

All the assigned Linux system administration and file management tasks were successfully executed on the Ubuntu Linux terminal. The MarketingAnalytics project directory structure was created, files were populated with sample data, searched, copied, renamed, moved, and deleted as required, demonstrating effective file and directory management for the Marketing Analytics team's project environment.

Key Questions / Analysis:

Q1. What is the difference between absolute and relative paths?

An absolute path specifies the complete location of a file or directory starting from the root directory (/), for example /home/user/MarketingAnalytics/Reports. It remains valid regardless of the current working directory. A relative path specifies the location of a file or directory relative to the current working directory, for example ./Reports or ../RawData. Relative paths are shorter but their meaning changes depending on where the user currently is in the file system.

Q2. Why are Linux permissions important for file security?

Linux permissions (read, write, execute) control which users and groups can access, modify, or execute a file or directory. They prevent unauthorized users from viewing sensitive data, altering critical files, or running unapproved scripts. In an enterprise environment like DataAnalytics Inc., permissions ensure that only authorized team members can access customer data and reports, protecting data confidentiality, integrity, and system security.

Q3. What is the purpose of the grep command?

The grep command is used to search for a specific pattern or text string within one or more files. It scans file contents line by line and displays the lines that match the given pattern. It is extremely useful for quickly locating specific information, such as searching for a customer name or keyword inside large log files or reports, without manually opening each file.

Q4. How does Linux support efficient file management?

Linux provides a hierarchical file system along with a rich set of command-line utilities (mkdir, cp, mv, rm, find, grep, ls, etc.) that allow users to create, organize, search, copy, move, and delete files and directories quickly. It supports wildcards, piping, redirection, and scripting, which enable automation of repetitive file management

tasks, making operations faster and more efficient compared to manual, GUI-based management, especially on servers.

Q5. Why are backups important in enterprise environments?

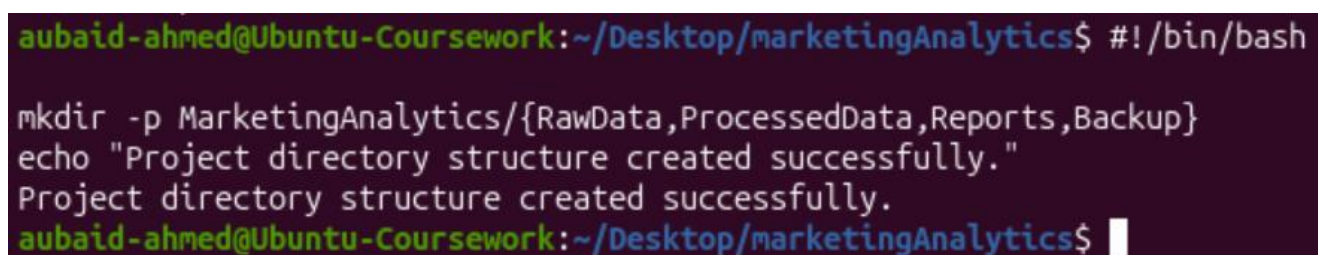
Backups protect an organization against data loss caused by hardware failure, accidental deletion, corruption, cyberattacks, or natural disasters. In an enterprise like DataAnalytics Inc., losing customer data or analytics reports could disrupt business operations and damage client trust. Regular backups ensure business continuity, allow quick recovery of critical data, and help the organization comply with data protection and audit requirements.

Supplementary Problems:

1. Create a shell script to automate project directory creation.

```
#!/bin/bash
mkdir -p MarketingAnalytics/{RawData,ProcessedData,Reports,Backup}
echo "Project directory structure created successfully."
```

Output:

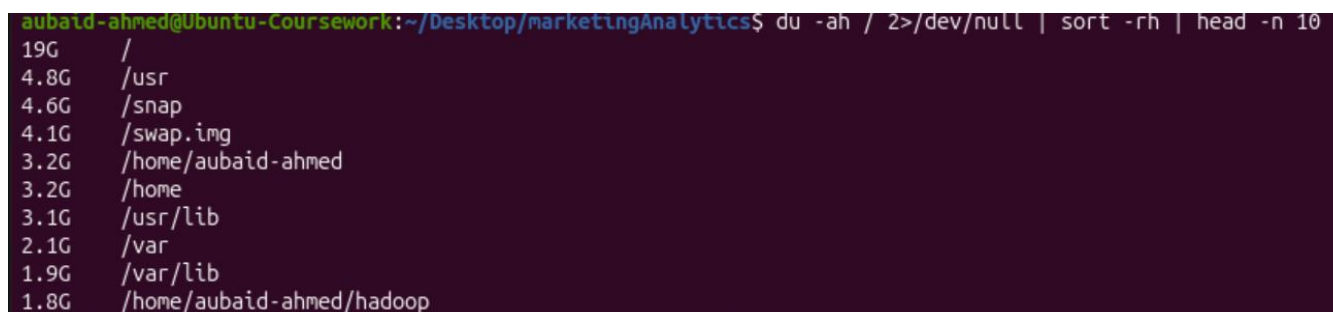


```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ #!/bin/bash
mkdir -p MarketingAnalytics/{RawData,ProcessedData,Reports,Backup}
echo "Project directory structure created successfully."
Project directory structure created successfully.
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$
```

2. Find the top 10 largest files in the system.

```
du -ah / 2>/dev/null | sort -rh | head -n 10
```

Output:



```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ du -ah / 2>/dev/null | sort -rh | head -n 10
19G      /
4.8G     /usr
4.6G     /snap
4.1G     /swap.img
3.2G     /home/aubaid-ahmed
3.2G     /home
3.1G     /usr/lib
2.1G     /var
1.9G     /var/lib
1.8G     /home/aubaid-ahmed/hadoop
```

3. Schedule a backup task using cron.

```
crontab -e
# Add the following line to run a backup daily at 11 PM:
0 23 * * * cp -r /home/user/MarketingAnalytics/Reports /home/user/MarketingAnalytics/Backup
```

Output:

```

aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ crontab -e
no crontab for aubaid-ahmed - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano        <---- easiest
 2. /usr/bin/vim.tiny
 3. /bin/ed

Choose 1-3 [1]: 0 23 * * * cp -r /home/user/MarketingAnalytics/Reports /home/user/MarketingAnalytics/Backup
Choose 1-3 [1]: ^O
Choose 1-3 [1]: ^X
Choose 1-3 [1]: 1
crontab: installing new crontab
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ crontab -l
0 23 * * * cp -r /home/user/MarketingAnalytics/Reports /home/user/MarketingAnalytics/Backup# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#

```

4. Monitor CPU utilization continuously using top/htop.

top

or, if installed:

htop

Output:

```

aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ top

top - 22:10:54 up 34 min,  1 user,  load average: 0.36, 0.45, 0.61
Tasks: 247 total,  1 running, 245 sleeping,  0 stopped,  1 zombie
%Cpu(s):  0.5 us,  0.7 sy,  0.0 ni, 98.4 id,  0.2 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :  6491.1 total,  825.3 free, 2821.6 used, 2948.5 buff/cache
MiB Swap:  4096.0 total,  4095.4 free,  0.6 used, 3669.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 1976 aubaid-+  20   0 5091964 379648 183140 S   5.2   5.7   3:45.50 gnome-shell
 3083 aubaid-+  20   0 2886184 429968 144312 S   2.1   6.5   1:10.05 ptxis
 3404 aubaid-+  20   0  11.7g 659604 265292 S   0.6   9.9   3:36.75 firefox
 4155 aubaid-+  20   0 2606248 96172  70336 S   0.6   1.4   0:06.78 Isolated Servic
  672 systemd+ 20   0  16808   7668   6624 S   0.3   0.1   0:01.11 systemd-oomd
 1130 root      20   0  347408 21436  18356 S   0.3   0.3   0:01.82 NetworkManager
 3595 aubaid-+  20   0 2697268 171428 110852 S   0.3   2.6   0:07.25 Privileged Cont
 4149 aubaid-+  20   0 3576392 925440 132328 S   0.3  13.9   2:14.29 Isolated Web Co
 4373 aubaid-+  20   0 2576732 73660  58264 S   0.3   1.1   0:01.46 Web Content
 5484 aubaid-+  20   0 22092  6332  4180 R   0.3   0.1   0:00.04 top
    1 root      20   0  25916  16824  11708 S   0.0   0.3   0:10.99 systemd
    2 root      20   0      0      0      0 S   0.0   0.0   0:00.09 kthreadd
    3 root      20   0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
   10 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-kblockd
   11 root      20   0      0      0      0 I   0.0   0.0   0:00.48 kworker/0:1-mm_percpu_wq
   12 root      20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/u16:0-ipv6_addrconf
   13 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root      20   0      0      0      0 S   0.0   0.0   0:00.28 ksoftirqd/0
   15 root      20   0      0      0      0 I   0.0   0.0   0:02.93 rcu_preempt
   16 root      20   0      0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0

```

5. Generate a disk usage report for all project directories.

du -sh MarketingAnalytics/*

Output:

```
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$ du -sh MarketingAnalytics/*
4.0K   MarketingAnalytics/Backup
4.0K   MarketingAnalytics/ProcessedData
4.0K   MarketingAnalytics/RawData
4.0K   MarketingAnalytics/Reports
aubaid-ahmed@Ubuntu-Coursework:~/Desktop/marketingAnalytics$
```

Tools / Technology used:

Ubuntu Linux / CentOS Linux, Linux Terminal, Bash Shell

Conclusion:

Through this practical, the fundamentals of Linux system administration and file management were learned and applied in a realistic business scenario. Commands for user identification, system monitoring, directory and file creation, content searching, and backup operations were practiced, giving hands-on exposure to how a Linux administrator manages a project environment such as the Marketing Analytics team's data, reports, and scripts. This forms a strong foundation for efficient, secure, and organized enterprise data management.